

Solution Architecture: shopez

Date: 26 June 2025

Team ID: LTVIP2025TMID55813

Project Name: shopez-one-stop-shop-for-online-purchases

Solution Architecture

Solution architecture is a structured process that connects business needs with technology solutions. In the shopez-one-stop-shop-for-online-purchases, the goal is to build a seamless online grocery shopping platform with secure user authentication, product filtering, order management, and a modern, user-friendly interface.

Goals of the Solution Architecture

-  Build a full-stack application using **MERN stack** (MongoDB, Express, React, Node.js).
-  Enable secure **user authentication** using JWT tokens.
-  Provide **category-wise filtering**, search, and product sorting.
-  Allow users to **add items to cart** and place orders.
-  Track and display **order history and order status**.
-  Build a visually rich UI with **product images, prices, brand names**, and smooth CSS styling.

Shopez Project Architecture and Data Flow

A typical architecture and data flow for the ShopSmart platform includes:

1. Frontend (React.js):

- User registration & login
- Product listing and filtering
- Cart and checkout functionality
- Order history and user profile

2. Backend (Node.js + Express):

- RESTful API endpoints for products, users, cart, orders
- Authentication and authorization (JWT)
- Business logic and route handling

3. **Database (MongoDB):**

- Collections: Users, Products, Orders, Cart Items
- Schema-based modeling with Mongoose

4. **Authentication (JWT):**

- Secure token-based login and protected routes
- Role-based access for admin and users

5. **Data Flow Overview:**

- User actions (e.g., add to cart) trigger frontend requests
- Frontend sends data to backend API
- Backend processes logic and updates the database
- Responses are sent back and rendered in the UI